

Part-Aware Embedding for Fine-Grained 3D Shape Retrieval

Suwoong Lee
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
Korea Advanced Institute of Science
and Technology (KAIST)
Daejeon, Republic of Korea
suwoong@etri.re.kr

Da-un Jung
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
crisia@etri.re.kr

Juwon Lee
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
zacurr@etri.re.kr

Seungjae Lee
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
seungjlee@etri.re.kr

Junmo Kim
Korea Advanced Institute of Science
and Technology (KAIST)
Daejeon, Republic of Korea
junmo.kim@kaist.ac.kr

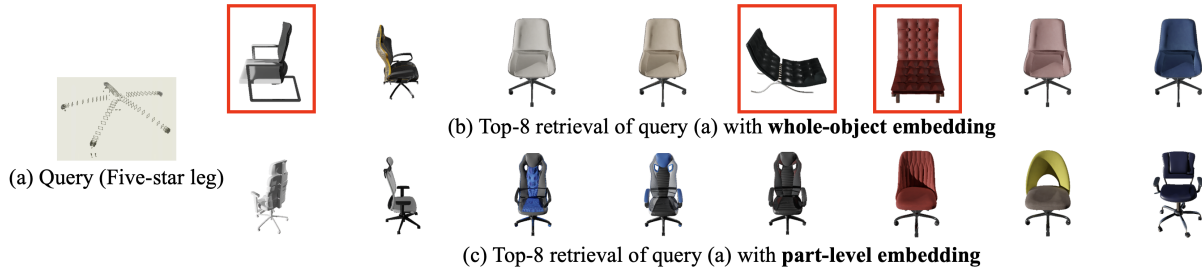


Figure 1: Illustration of retrieval. (a) The query shape. (b) Top-8 retrieval results using whole-object embeddings [Liu et al. 2023]. (c) Top-8 retrieval results using part-level embeddings (ours). Objects with red box represent the mis-retrieved results.

Abstract

We propose a part-aware retrieval framework for 3D shapes that enables fine-grained similarity analysis by embedding both whole objects and their constituent parts into a unified feature space. Unlike previous methods that rely on global object-level embeddings and often overlook locally distinctive structures, our approach incorporates part segmentation and part-level embeddings to enhance retrieval accuracy, particularly for queries targeting specific components. Evaluations on a custom dataset derived from Objaverse V1 demonstrate that our method outperforms whole-object-only retrieval in both quantitative metrics and qualitative results.

ACM Reference Format:

Suwoong Lee, Da-un Jung, Juwon Lee, Seungjae Lee, and Junmo Kim. 2025. Part-Aware Embedding for Fine-Grained 3D Shape Retrieval. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771499>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771499>

1 Introduction

Recent advances in 3D shape retrieval have employed image-text foundation models to align multimodal data—such as images, text, and 3D shapes—into a shared embedding space [Liu et al. 2023; Schlachter et al. 2022]. This alignment enables retrieval via multi-modal queries. However, most existing methods generate a *single embedding per 3D object*, which limits their ability to represent and compare localized shape features, especially in complex objects.

To address this limitation, we propose a retrieval framework that embeds both whole objects and their constituent parts into a shared metric space. By using part-aware embeddings, our method enables queries to be matched directly to geometrically similar parts within the database, rather than relying solely on whole-object similarity.

As shown in Figure 1, a query based on a five-star leg part retrieves different results depending on whether whole-object or part-level embeddings are used. Part-level embedding retrieves objects that have a five-star leg structure, while whole-object embedding retrieves almost random leg structures. This demonstrates that part-aware representations allow more accurate and fine-grained retrieval when local shape details are critical.

2 Proposed Method

Previous 3D shape retrieval methods typically measure similarity using the cosine distance between global object-level embeddings,

which we refer to as the whole-object distance:

$$d_{\text{whole}} = 1 - \frac{v_q \cdot v_o}{\|v_q\| \|v_o\|} \quad (1)$$

where v_q is the embedding of the query and v_o is the embedding of the target object. While this approach is effective for capturing overall shape similarity, it often fails to distinguish local geometric differences, especially when the query focuses on a specific part.

To overcome this limitation, we propose a part-aware retrieval framework that embeds both whole objects and their constituent parts into a shared metric space. By using part-aware embeddings, our method enables queries to be matched directly to geometrically similar parts within the database, rather than relying solely on whole-object similarity. Specifically, we define the part-aware distance as the minimum distance between the query embedding and the part embeddings of the target object:

$$d_{\text{part}} = \min_{p \in P} \left(1 - \frac{v_q \cdot v_{o_p}}{\|v_q\| \|v_{o_p}\|} \right) \quad (2)$$

where P is the set of parts of the target object and v_{o_p} is the embedding of part p . This enables the retrieval system to directly match the query to semantically and geometrically similar parts, resulting in more fine-grained and accurate retrieval, particularly for queries targeting local object features.

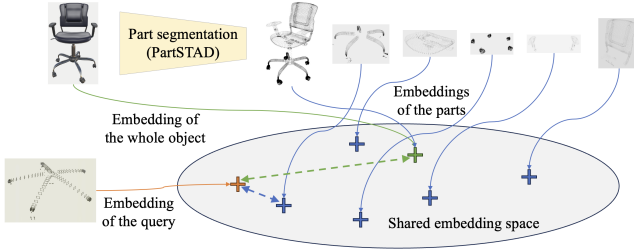


Figure 2: A conceptual diagram of measuring the distance between a query and a target object. The green and blue dashed lines represent the whole-object distance and the proposed part-aware distance, respectively.

Figure 2 illustrates the difference between the previous whole-object distance (Eq. 1) and proposed part-aware distance (Eq. 2).

By leveraging semantically similar parts (e.g., five-star leg shapes), our approach enables fine-grained, part-based 3D shape retrieval.

For part segmentation, we adopt the PartSTAD (Part Segmentation Task Adaptation) framework [Kim and Sung 2024], which utilizes a PartNet-Ensemble model pre-trained on the PartNet-Mobility dataset. The input point cloud is projected into 2D images from multiple viewpoints, and 2D bounding boxes for each part are predicted using GLIP based on text prompts. SAM is then applied to segment the identified regions and generate 2D masks, which are mapped back onto the 3D point cloud.

After segmenting the database object into parts and computing their embeddings, the system calculates the part-aware distance between the query and each database object. Retrieval results are obtained by ranking objects according to this distance, with top-K results returned to the user. This pipeline allows our method to retrieve objects based on fine-grained, part-level similarity robustly.

3 Experimental Results

To evaluate our method, we construct custom datasets for chairs and lamps based on Objaverse V1 [Deitke et al. 2023]. Each object is segmented into semantic parts and annotated with category-specific labels (e.g., leg type or shade style) for fine-grained retrieval. We compute embeddings using a PointBert-based encoder aligned with the OpenCLIP ViT-G/14 model, following the OpenShape [Liu et al. 2023] framework. Details of the experimental setup are provided in the Appendix.

To evaluate the retrieval performance, we compute the precision@K between whole-object and part-level embeddings on both the chair and lamp datasets. Precision@K is the percentage of correct retrieval results among the top K retrieved results. Table 1 and Table 2 show the results of precision@K between whole-object and part-level embeddings on the chair and lamp dataset, respectively. The part-level retrieval consistently achieves higher precision among all K values. More quantitative evaluations and qualitative evaluations are provided in the Appendix.

Table 1: Comparison of Precision@K for whole-object and part-level embeddings on the chair dataset. The best average performance is underlined.

Embedding	Class	P@1	P@3	P@5	P@10
Whole-object level [Liu et al. 2023]	straight leg	0.167	0.222	0.222	0.333
	five-star leg	0.966	0.908	0.876	0.845
	average	<u>0.566</u>	<u>0.565</u>	<u>0.549</u>	<u>0.589</u>
Part level (Ours)	straight leg	0.889	0.926	0.922	0.900
	five-star leg	1.000	1.000	1.000	0.986
	average	<u>0.944</u>	<u>0.963</u>	<u>0.961</u>	<u>0.943</u>

Table 2: Comparison of Precision@K for whole-object and part-level embeddings on the lamp dataset. The best average performance is underlined.

Embedding	Class	P@1	P@3	P@5	P@10
Whole-object level [Liu et al. 2023]	bell	0.400	0.233	0.200	0.190
	bowl	0.545	0.515	0.491	0.468
	drum	0.618	0.451	0.482	0.491
	empire	0.333	0.347	0.325	0.288
	average	<u>0.474</u>	<u>0.387</u>	<u>0.375</u>	<u>0.359</u>
Part level (Ours)	bell	0.200	0.200	0.180	0.160
	bowl	0.750	0.667	0.618	0.623
	drum	0.824	0.637	0.588	0.556
	empire	0.458	0.403	0.425	0.388
	average	<u>0.558</u>	<u>0.477</u>	<u>0.453</u>	<u>0.432</u>

4 Conclusion

We introduced a part-aware 3D shape retrieval framework that aligns part and object representations in a shared feature space. Our method allows users to retrieve objects based on localized geometric features, addressing the limitations of whole-object-only approaches. Evaluation on a custom Objaverse V1-based dataset shows that our approach supports more discriminative retrieval.

Acknowledgments

This work was supported by internal fund/grant of ETRI (Project No.: 24RC1500, Contribution Rate:50%) and 2025 Cultural Heritage Smart Preservation & Utilization R&D Program of Korea Heritage Service, National Research Institute of Cultural Heritage (Project No.: RS-2024-00396158, Contribution Rate:50%)

References

- Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli Vander-Bilt, Ludwig Schmidt, Kiana Ehsani, Aniruddha Kembhavi, and Ali Farhadi. 2023. Objaverse: A Universe of Annotated 3D Objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 13142–13153.
- Hyunjin Kim and Minhyuk Sung. 2024. PartSTAD: 2D-to-3D Part Segmentation Task Adaptation. In *European Conference on Computer Vision*. Springer, 422–439.
- Minghua Liu, Ruoxi Shi, Kaiming Kuang, Yinhao Zhu, Xuanlin Li, Shizhong Han, Hong Cai, Fatih Porikli, and Hao Su. 2023. OpenShape: Scaling Up 3D Shape Representation Towards Open-World Understanding. *Advances in neural information processing systems* 36 (2023), 44860–44879.
- Kristofer Schlachter, Benjamin Ahlbrand, Zhu Wang, Ken Perlin, and Valerio Ortenzi. 2022. Zero-Shot Multi-Modal Artist-Controlled Retrieval and Exploration of 3D Object Sets. In *SIGGRAPH Asia 2022 Technical Communications*. 1–4.